

Customer Shopping Behavior Analysis Project

EDA · Python | SQL · PostgreSQL | Dashboard · Power BI

Data Analysis End-to-End

What's Inside This Presentation

01

Project Overview & Dataset

02

Python EDA – Exploratory Data Analysis

03

SQL – Business Problem Solving

04

Power BI Dashboard

05

Key Insights & Recommendations

Project Overview

End-to-End Data Analysis Portfolio Project Objective

To analyze customer shopping behavior, identify key revenue patterns, and deliver actionable business insights using Python, SQL and Power BI.

5,050

Total Records

17

Columns

5

Categories

12

SQL Queries

Phase 1 – Python EDA

Import data → Clean nulls → Fix categories →
Remove duplicates → Export to PostgreSQL

Phase 2 – SQL Analysis

12 business questions solved using SELECT,
GROUP BY, CASE, CTEs, Window Functions

Phase 3 – Power BI

Interactive dashboard with slicers for revenue,
category, gender, subscription & shipping
analysis

Dataset Overview

customer_shopping_behavior.csv — 5,050 rows × 17 columns

Key Columns:

Customer ID	Size	Previous Purchases
Age	Color	Payment Method
Gender	Season	Frequency of Purchases
Item Purchased	Review Rating	
Category	Subscription Status	
Purchase Amount (USD)	Shipping Type	
Location	Discount Applied	

Null Values Found Before Cleaning:

601	556	548	370
Review Rating	Purchase Amount (USD)	Previous Purchases	Size

SECTION 01

Python EDA

Exploratory Data Analysis using Pandas

EDA — Step-by-Step Process

Python · Jupyter Notebook · Pandas Library

1

Data Import

Imported dataset using `pd.read_csv()`. Verified shape: 5050 rows × 17 columns.

2

Data Exploration

Used `df.head()`, `df.tail()`, `df.info()`, `df.describe(include='all')` to understand structure and stats.

3

Category Correction

Items were mapped to wrong categories. Created a `correct_mapping` dictionary to fix all 5 categories: Clothing, Electronics, Accessories, Footwear, Outerwear.

4

Null Value Treatment

Size: Set 'Not Applicable' for Electronics, 'Free Size' for Accessories, mode for Clothing.
Review Rating: Filled with per-item mean.
Purchase Amount: Filled with per-item mean.
Previous Purchases: Filled with 0.

5

Duplicate Removal

Detected and removed duplicate Customer IDs using `df.drop_duplicates(subset=['Customer ID'], keep='first')`.

6

Column Standardization

Renamed columns: spaces → underscores, all lowercase. Renamed 'purchase_amount_(usd)' to 'purchase_amount'.

EDA — Key Findings & Output

After data cleaning: 5,000 clean records ready for SQL analysis

Before & After Cleaning

Metric	Before Cleaning	After Cleaning	Improvement
Total Rows	5,050	5,000	50 duplicates removed
Null Values	2,075	0	All nulls treated
Categories	Mixed/Wrong	5 Clean	Category mapping fixed
Column Names	Spaces, Mixed Case	snake_case	Standardized

Data Export to PostgreSQL

```
from sqlalchemy import create_engine
engine = create_engine("postgresql+psycopg2://postgres:****@localhost:5433/customer_shopping_behavior")
df.to_sql(name="customer", con=engine, if_exists="replace", index=False, chunksize=500)
# ✅ 'customer' table created and data inserted successfully!
```

SECTION 02

SQL Analysis

12 Business Problems Solved in PostgreSQL

SQL Business Queries — Part 1

PostgreSQL · Business Problem → Query → Output → Impact

Q1 Which category generates highest revenue?

Problem: We do not know which category gives the most revenue.

```
SELECT category,
       ROUND(SUM(purchase_amount::NUMERIC),2) AS highest_revenue
FROM customer
GROUP BY category
ORDER BY highest_revenue DESC;
```

Impact:

Helps identify top revenue categories · Supports inventory decisions · Improves marketing strategy

 **Result:** Electronics: \$486,367 | Accessories: \$194,867 | Clothing: \$183,188 | Footwear: \$111,236 | Outerwear: \$18,524

Q2 Are discounts actually increasing purchase value?

Problem: We do not know if discounts are helping or reducing profit.

```
SELECT discount_applied,
       ROUND(SUM(purchase_amount::NUMERIC),2) AS highest_revenue,
       ROUND(AVG(purchase_amount::NUMERIC),2) AS avg_purchase
FROM customer
GROUP BY discount_applied;
```

Impact:

Measures real impact of discounts · Avoids unnecessary discounts · Improves profit margin

 **Result:** No Discount: Sum=\$502,242 | Avg=\$182 · With Discount: Sum=\$491,941 | Avg=\$219 → Discounts increase avg spend by \$37

SQL Business Queries — Part 2

PostgreSQL · Business Problem → Query → Output → Impact

Q3 What is the total revenue by gender?

Problem: We do not know which gender spends more.

```
SELECT gender,
       ROUND(SUM(purchase_amount::NUMERIC),2) AS highest_revenue
FROM customer
GROUP BY gender
ORDER BY highest_revenue DESC;
```

Impact:

Understand buying patterns · Support targeted campaigns · Improve personalization

 **Result:** Male: \$390,965 | Female: \$352,060 | Other: \$251,158 → Male customers generate the most revenue

Q4 High-value discount customers (above avg spend)

Problem: We cannot find customers who use discounts but still spend more.

```
SELECT customer_id, purchase_amount, discount_applied
FROM customer
WHERE discount_applied = 'Yes'
      AND purchase_amount > (SELECT AVG(purchase_amount) FROM customer)
LIMIT 10;
```

Impact:

Identify high-value customers · Smarter discount targeting · Increase retention

 **Result:** 10 sample customers found — IDs like 4568, 4256, 4289 — all with purchase amounts above \$276 while using discounts

SQL Business Queries — Part 3

PostgreSQL · Business Problem → Query → Output → Impact

Q5 Top & Bottom 5 products by average review rating

Problem: We do not know which products customers like or dislike.

```
-- Top 5
SELECT item_purchased, ROUND(AVG(review_rating::NUMERIC),2) AS avg_rating
FROM customer GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC LIMIT 5;
```

Impact:

Identify top/low performing products · Support product improvement · Enhance satisfaction

 **Result:** Top 5: Gloves(3.86), Sandals(3.84), Boots(3.82), Hat(3.80), Skirt(3.79)
Bottom 5: Phone(2.94), Headphones(3.08), Watch(3.10), Bag(3.17), Laptop(3.19)

Q6 Average purchase by shipping type

Problem: We do not know if fast shipping increases spending.

```
SELECT shipping_type,
COUNT(DISTINCT customer_id) AS order_placed,
ROUND(AVG(purchase_amount::NUMERIC),2) AS avg_purchase,
ROUND(SUM(purchase_amount::NUMERIC),2) AS revenue
FROM customer
GROUP BY shipping_type ORDER BY revenue DESC;
```

Impact:

Evaluate shipping impact on sales · Better delivery pricing · Encourage higher order value

 **Result:** Express: \$429,381 (avg \$358) · Standard: \$409,020 (avg \$341) · Free Shipping: \$40,777 (avg \$60) · Store Pickup: \$38,931

SQL Business Queries — Part 4

PostgreSQL · Business Problem → Query → Output → Impact

Q7 Do subscribed customers spend more?

Problem: We do not know if subscription customers spend more.

```
SELECT subscription_status,
       COUNT(customer_id) AS users,
       ROUND(AVG(purchase_amount::NUMERIC),2) AS avg_revenue,
       ROUND(SUM(purchase_amount::NUMERIC),2) AS total_revenue
FROM customer
GROUP BY subscription_status ORDER BY total_revenue DESC;
```

Impact:

Evaluate subscription model · Improve loyalty programs · Increase long-term customer value

 **Result:** Non-Subscribers: 3,421 users | Avg \$166 | Total \$566,937
Subscribers: 1,579 users | Avg \$271 | Total \$427,245 → Subscribers spend 63% MORE on average

Q8 Top 5 products with highest discount usage %

Problem: Some products depend too much on discounts.

```
SELECT item_purchased,
       COUNT(item_purchased) AS total_sold,
       COUNT(CASE WHEN discount_applied='Yes' THEN 1 END) AS discount_count,
       COUNT(CASE WHEN discount_applied='Yes' THEN 1 END)*100.0/COUNT(*) AS
discount_pct
FROM customer
GROUP BY item_purchased ORDER BY discount_pct DESC LIMIT 5;
```

Impact:

Identify discount-driven products · Better pricing strategy · Reduce margin loss

 **Result:** Laptop: 51.7% · Phone: 51.5% · Headphones: 51.2% · Hat: 50% · Watch: 49.7% → Electronics are most discount-dependent

SQL Business Queries — Part 5

PostgreSQL · Business Problem → Query → Output → Impact

Q9

Customer segmentation: New, Returning, Loyal

Problem: We are not grouping customers properly.

```
SELECT
  CASE WHEN previous_purchases = '0' THEN 'New Customer'
        WHEN previous_purchases BETWEEN 1 AND 15 THEN 'Returning Customer'
        ELSE 'Loyal Customers' END AS customer_segment,
  COUNT(*) AS customer_count
FROM customer GROUP BY 1;
```

Impact:

Enable clear segmentation · Personalized marketing · Improve retention

 **Result:** Loyal Customers: 3,085 (61.7%) · Returning Customers: 1,354 (27.1%) · New Customers: 561 (11.2%)

Q10

Top 3 most purchased products within each category

Problem: We do not know top products in each category.

```
WITH CTE AS (
  SELECT category, item_purchased,
         COUNT(item_purchased) AS most_purchased,
         RANK() OVER (PARTITION BY category ORDER BY COUNT(item_purchased) DESC) AS RNK
  FROM customer GROUP BY category, item_purchased
)
SELECT * FROM CTE WHERE RNK <= 3;
```

Impact:

Identify best-sellers · Better product placement · Focused promotions

 **Result:** Accessories: Jewelry(171), Bag(164), Belt(161) · Clothing: Shirt(315), Pants(171) · Electronics: Phone(171), Headphones(166) · Footwear: Shoes(303)

SQL Business Queries — Part 6

PostgreSQL · Business Problem → Query → Output → Impact

Q11 Are repeat buyers likely to subscribe?

Problem: We do not know if loyal customers take subscriptions.

```
SELECT
  CASE WHEN previous_purchases > 5 THEN 'Repeat Buyers' ELSE 'Normal Buyers' END
AS type,
  subscription_status, COUNT(*) AS count,
  COUNT(*)*100.0 / SUM(COUNT(*)) OVER (PARTITION BY ...) AS pct
FROM customer GROUP BY 1, 2;
```

Impact:

Understand loyalty behavior · Better subscription targeting · Improve conversion

 **Result:** Normal Buyers: 37% subscribed · Repeat Buyers: 30% subscribed → Repeat buyers are NOT more likely to subscribe — opportunity to upsell

Q12 Revenue contribution of each age group

Problem: We do not know which age group spends more.

```
SELECT
  CASE WHEN age BETWEEN 18 AND 25 THEN '18-25'
        WHEN age BETWEEN 26 AND 35 THEN '26-35'
        WHEN age BETWEEN 36 AND 50 THEN '36-50'
        ELSE '51+' END AS age_group,
  ROUND(SUM(purchase_amount::NUMERIC),2) AS total_revenue
FROM customer GROUP BY 1 ORDER BY total_revenue DESC;
```

Impact:

Identify high-value age groups · Targeted marketing by age · Improve campaign effectiveness

 **Result:** 51+ Age Group: \$399,345 (HIGHEST) · 36-50: \$266,707 · 26-35: \$177,422 · 18-25: \$150,708 → Older customers are top spenders

SECTION 03

Power BI Dashboard

Interactive Visual Analytics & Business Reporting



CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Category

All

Gender

Female

Male

Other

Subscription Status

No

Yes

Discount Applied

No

Yes



5K

Total Customers



30

Unique Items



\$994.2K

Total Revenue



\$198.8

Avg Order Value

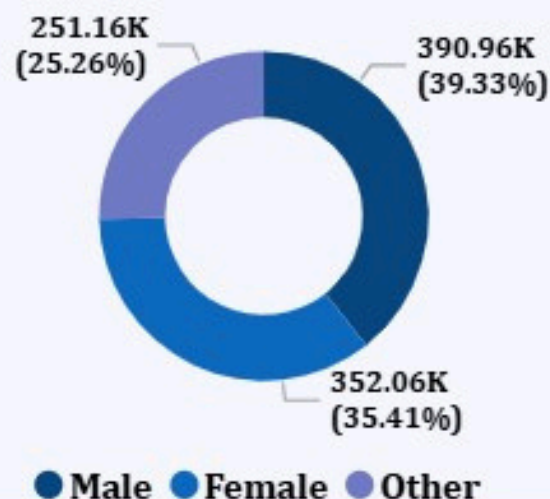


3.6

Average Review



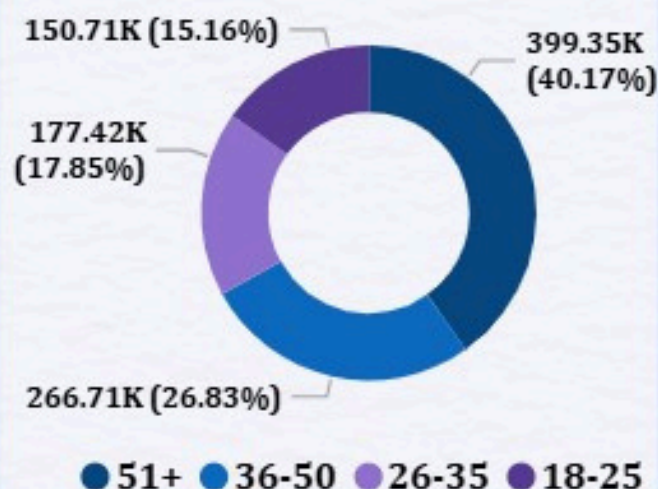
Revenue by Gender



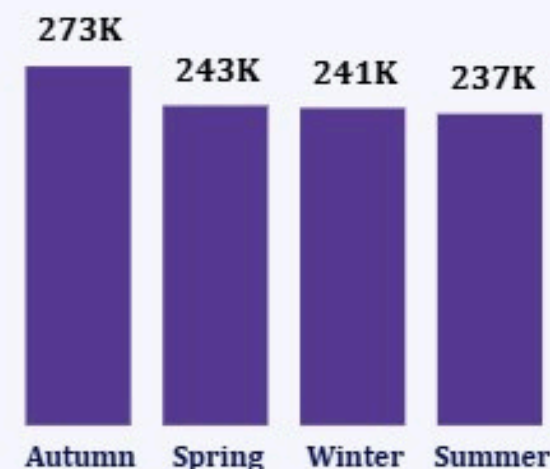
Revenue by Category



Revenue by Age Group



Revenue by Season



Key Insights

- Gender**
 - Male customers contribute the highest share of revenue, followed by Female customers, while Other customers contribute the least
- Category**
 - Electronics category generates the highest revenue, significantly outperforming all other categories
- Age Group**
 - Customers aged 51+ contribute the highest share of revenue, indicating strong purchasing power compared to other age groups
- Season**
 - Autumn shows the highest revenue, while Spring, Winter, and Summer maintain consistently strong performance.
- Location**
 - New York shows the highest average order value, indicating high-value customer purchases, while smaller locations contribute less to overall revenue
- Shipping Type**
 - Standard and Express shipping types have the highest number of orders,

Orders and Average Spend by Location



Orders and Average Spend by Shipping Type



Revenue by Payment Method



Customer Behavior & Revenue Analysis

Best vs Worst Performance





CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Category

All

Gender

Female

Male

Other

Subscription Status

No

Yes

Discount Applied

No

Yes



5K

Total Customers



30

Unique Items



\$994.2K

Total Revenue



\$198.8

Avg Order Value



3.6

Average Review



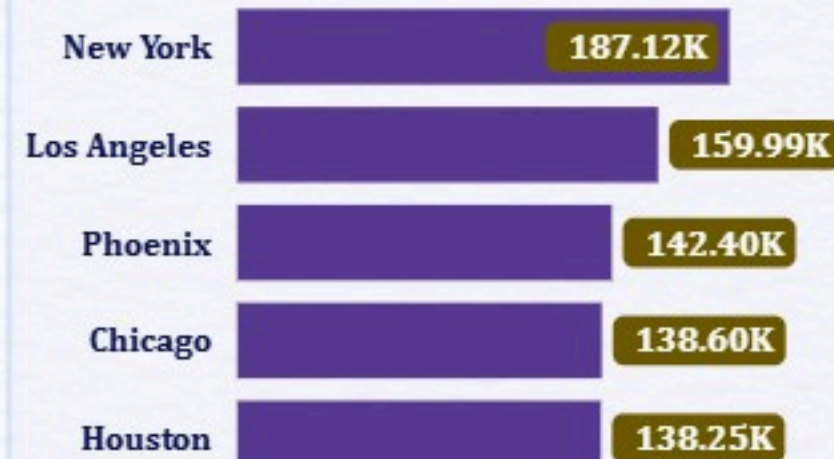
Top 5 Products by Rating



Top 5 Products by Revenue



Top 5 Location by Revenue



Key Insights

1 Rating

Jewelry is the **highest rated** item, while **Phone** has the **lowest rating**. **Headphones** generates the **most revenue**, while **Jeans** generates the **least revenue**.

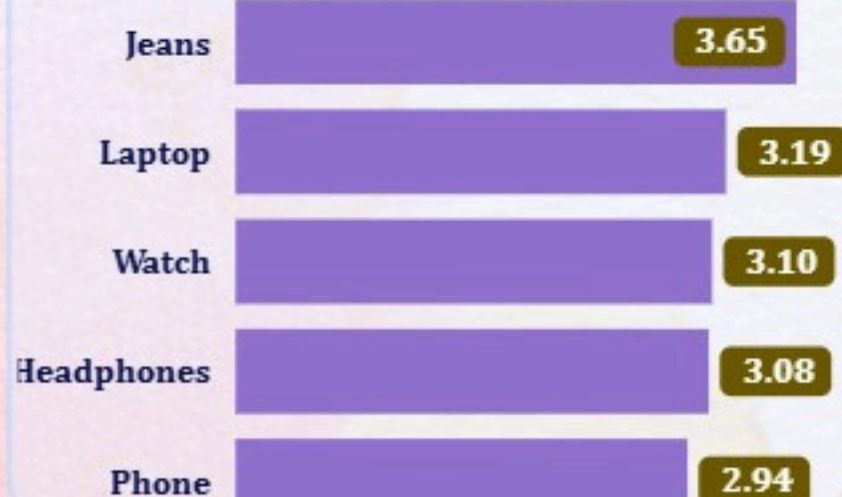
2 Revenue

Electronics items like **Laptop**, **Watch**, **Phone** & **Headphones** appear in **both top revenue AND bottom ratings lists** — they sell well but customers are not satisfied.

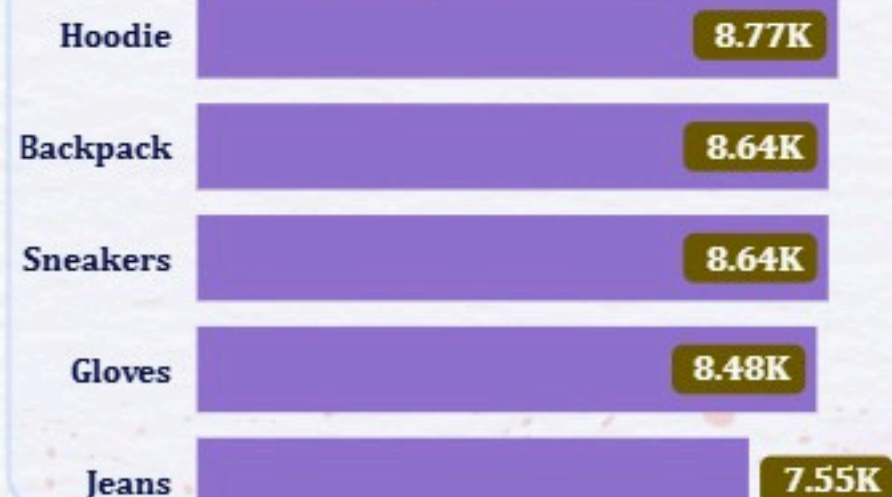
3 Location

New York is the **top performing city** with **\$187K** in revenue. **Kansas** is the **lowest performing city** with only **\$3.44K** in revenue.

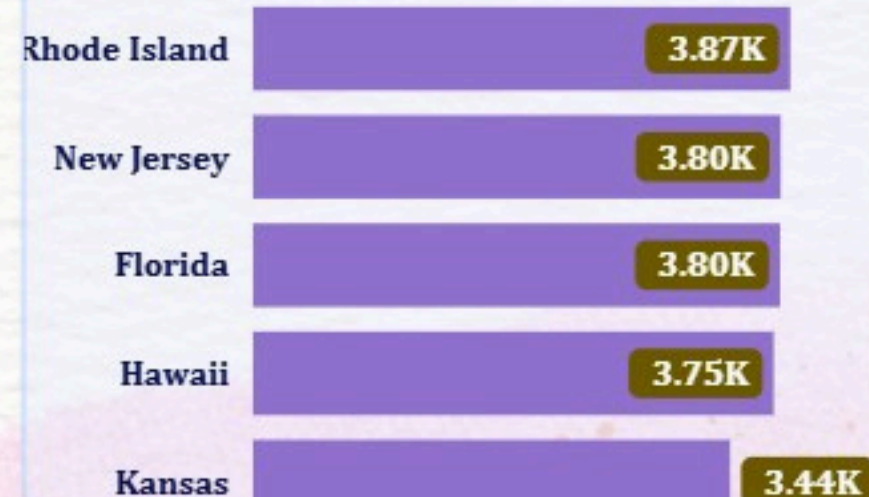
Bottom 5 Products by Rating



Bottom 5 Products by Revenue



Bottom 5 Location by Revenue



Power BI Dashboard — Overview

Interactive dashboard built on cleaned data from Python EDA

Dashboard Highlights

Revenue by Category

Bar chart showing Electronics leads with \$486K. Easy to compare all 5 categories.

Gender-wise Revenue

Pie/donut chart. Male: \$391K, Female: \$352K, Other: \$251K. Helps in targeted marketing.

Discount Impact

Visual showing customers WITH discount have higher avg spend (\$219 vs \$182). Discount IS beneficial.

Subscription Analysis

Subscribers spend 63% more on average (\$271 vs \$166). Subscription model is profitable.

Shipping Type Revenue

Express shipping generates \$429K — highest revenue. Standard also strong at \$409K.

Age Group Revenue

51+ age group contributes the most revenue (\$399K). Older customers = higher value buyers.

Power BI — Charts & Slicers Used

Interactive elements that make the dashboard dynamic

Visual Elements in Dashboard

KPI Cards

Total Revenue, Avg Purchase Amount, Total Customers, Avg Review Rating shown as big numbers at top

Bar Chart

Revenue by Category — clear visual comparison of all 5 product categories

Donut Chart

Revenue split by Gender — Male, Female, Other with % labels

Column Chart

Revenue by Age Group — 4 age bands side by side

Stacked Bar

Shipping Type vs Revenue — shows Express dominance clearly

Clustered Bar

Subscription Status vs Avg Purchase — clear comparison

Slicers

Filters for: Category, Gender, Discount Applied, Season, Subscription Status — makes it interactive

Table Visual

Top products with purchase count and review rating for drill-down analysis

SECTION 04

Key Insights

What the Data Tells Us — Actionable Business Findings

Key Insights from the Analysis

Consolidated findings across EDA, SQL and Power BI

Electronics = #1 Revenue Driver

Electronics alone generates \$486K — 2.5x more than the next category (Accessories at \$195K). Invest heavily here.

Discounts Work — But Selectively

Customers using discounts spend \$37 MORE on average (\$219 vs \$182). However, Laptop, Phone & Headphones have 50%+ discount dependency — risky for margins.

Subscribers Are Premium Customers

Only 1,579 subscribers but they generate \$427K total with avg spend of \$271 — 63% higher than non-subscribers (\$166 avg).

Express Shipping = More Revenue

Express shipping customers generate \$429K at avg \$358 per order. Standard shipping is \$341. Faster shipping correlates with bigger purchases.

51+ Age Group Dominates Revenue

Customers aged 51+ spend the most — contributing \$399K. Brands should create products and experiences for older, high-spending customers.

Loyal Customers (61%) Need Subscriptions

3,085 loyal customers exist but only 30% of repeat buyers are subscribed. Huge opportunity to convert loyal buyers into subscribers.

Business Recommendations

Actionable steps based on data insights

01

HIGH

Boost Electronics Inventory

Electronics is the #1 revenue category. Increase stock, improve product variety, and run targeted promotions.

02

HIGH

Optimize Discount Strategy

Apply discounts only on high-margin products. Avoid heavy discounting on Laptops, Phones & Headphones which are already 50%+ discount-dependent.

03

HIGH

Convert Loyal Buyers to Subscribers

Only 30% of repeat buyers subscribe. Launch a loyalty-to-subscription campaign with exclusive benefits to increase this number.

04

MEDIUM

Target 51+ Age Group

51+ customers are the top spenders. Create dedicated marketing campaigns, product bundles, and offers for this age segment.

05

MEDIUM

Promote Express Shipping

Express shipping users spend \$17 more per order on average. Offer free express shipping above a threshold to increase average order value.

06

MEDIUM

Improve Low-Rated Products

Phone (2.94 avg rating) and Headphones (3.08) have lowest ratings. Investigate quality issues and improve these products to boost customer satisfaction.

Tools & Technologies Used

Complete end-to-end data analysis tech stack



Python

Pandas, SQLAlchemy

Data cleaning, EDA, null treatment, duplicate removal, category correction and export to DB



Jupyter Notebook

IPython, ipykernel

Interactive environment for step-by-step EDA with code, output and markdown documentation



PostgreSQL (pgAdmin)

psycopg2

SQL database to store cleaned data and solve 12 real business problems using advanced queries



Power BI Desktop

DAX

Interactive dashboard with KPI cards, bar/donut/column charts, slicers and cross-filtering

Project Summary

Customer Shopping Behavior Analysis — Complete

5,000

Clean Records

17

Data Columns

12

SQL Queries

6

Key Insights

Python EDA → PostgreSQL SQL → Power BI Dashboard

Thank You for Viewing My Project!